

# Customer Shopping Behavior Analysis

## Project Report & Business Recommendations

Prepared for Stakeholder Review

### Executive Summary

This report summarizes an analysis of 3,900 customer transactions across 18 attributes, including demographics, product category, spending, discounts, shipping, and subscription status. The goal was to uncover the factors that drive consumer purchasing decisions and repeat business, and to translate those findings into actionable recommendations.

The analysis shows a loyal, engaged customer base (80% classified as "Loyal") but a very low rate of new customer acquisition (2%), a subscription program that is not translating into higher average spend, and a small group of high-discount product lines that may be eroding margin. Full findings and recommendations follow below.

### Headline Metrics

|                 |                         |                       |
|-----------------|-------------------------|-----------------------|
| 3,900           | \$59.76                 | 3.75 / 5              |
| Total Customers | Average Purchase Amount | Average Review Rating |

### Problem Statement

A leading retail company wants to better understand its customers' shopping behavior in order to improve sales, customer satisfaction, and long-term loyalty. Management has noticed changes in purchasing patterns across demographics, product categories, and sales channels, and wants to know which factors — discounts, reviews, seasonality, or payment preferences — drive consumer decisions and repeat purchases.

*Overarching business question: “How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?”*

### Methodology

The customer\_shopping\_behavior dataset (3,900 records, 18 fields) was loaded and cleaned in Python (missing review ratings imputed by category median; column names standardized; a redundant promo-code column removed after confirming it was identical to the discount-applied field). Ten targeted business questions were then answered using SQL queries against a PostgreSQL database, and the key metrics were consolidated into an interactive Power BI dashboard for ongoing monitoring.

### Key Findings

## 1. Revenue Skews Male, but Spend Per Customer Is Nearly Equal

| Gender | Revenue   |
|--------|-----------|
| Female | \$75,191  |
| Male   | \$157,890 |

Male customers make up roughly two-thirds of the customer base, which explains most of the revenue gap — average spend per customer is nearly identical between genders. The opportunity is customer count, not spending behavior.

## 2. Discount Usage Doesn't Always Mean Lower-Value Customers

A meaningful subset of customers who applied a discount still spent above the dataset's average purchase amount, confirming that discount usage and high-value spending are not mutually exclusive.

## 3. Highest-Rated Products

| Product | Avg. Rating |
|---------|-------------|
| Gloves  | 3.86        |
| Sandals | 3.84        |
| Boots   | 3.82        |
| Hat     | 3.80        |
| Skirt   | 3.78        |

## 4. Express Shipping Correlates with Higher Order Value

| Shipping Type | Avg. Purchase Amount |
|---------------|----------------------|
| Standard      | \$58.46              |
| Express       | \$60.48              |

## 5. Subscription Status Has Little Effect on Average Spend

| Status         | Customers | Avg. Spend | Total Revenue |
|----------------|-----------|------------|---------------|
| Subscribed     | 1,053     | \$59.49    | \$62,645      |
| Not Subscribed | 2,847     | \$59.87    | \$170,436     |

Subscribers and non-subscribers spend almost the same amount per order. The subscription program is not currently a lever for larger basket sizes.

## 6. Products Most Reliant on Discounts

| Product | % of Purchases Discounted |
|---------|---------------------------|
| Hat     | 50%                       |

| Product  | % of Purchases Discounted |
|----------|---------------------------|
| Sneakers | 49%                       |
| Coat     | 49%                       |
| Sweater  | 48%                       |
| Pants    | 47%                       |

## 7. Customer Base Is Loyalty-Heavy, Acquisition-Light

| Segment                          | Number of Customers | Share |
|----------------------------------|---------------------|-------|
| Loyal (11+ prior purchases)      | 3,116               | 80%   |
| Returning (2–10 prior purchases) | 701                 | 18%   |
| New (1 prior purchase)           | 83                  | 2%    |

Retention is clearly strong, but new customer inflow is minimal — a risk to long-term growth if not addressed.

## 8. Best-Selling Products by Category

| Category    | Top Products (by orders)                    |
|-------------|---------------------------------------------|
| Accessories | Jewelry (171), Sunglasses (161), Belt (161) |
| Clothing    | Blouse (171), Pants (171), Shirt (169)      |
| Footwear    | Sandals (160), Shoes (150), Sneakers (145)  |
| Outerwear   | Jacket (163), Coat (161)                    |

## 9. Most Repeat Buyers Are Not Subscribers

| Subscribed? | Repeat Buyers (>5 purchases) |
|-------------|------------------------------|
| No          | 2,518                        |
| Yes         | 958                          |

72% of the most loyal repeat buyers have not joined the subscription program — a clear conversion opportunity.

## 10. Revenue by Age Group

| Age Group   | Total Revenue |
|-------------|---------------|
| Young Adult | \$62,143      |
| Middle-Aged | \$59,197      |
| Adult       | \$55,978      |
| Senior      | \$55,763      |

Revenue is fairly evenly distributed across age groups, with Young Adults contributing slightly more than other segments.

## Power BI Dashboard

The metrics above were consolidated into an interactive dashboard for ongoing monitoring by category, age group, gender, and subscription status.

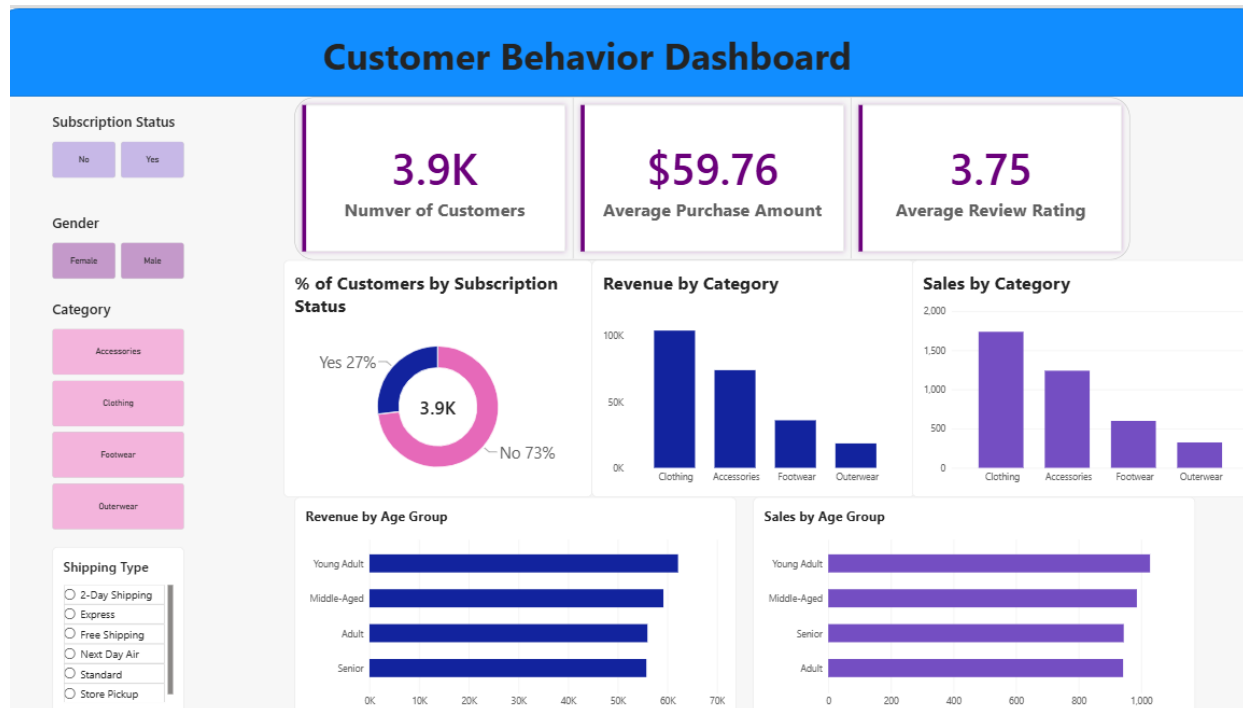


Figure 1. Customer Behavior Dashboard — key metrics by category, age group, gender, and subscription status.

## Business Recommendations

### 1. Grow the female customer base

- Spend per customer is comparable across genders, so the revenue gap is driven by customer count. Run targeted acquisition campaigns and category assortment reviews aimed at female shoppers to close the gap.

### 2. Prioritize new customer acquisition

- Only 2% of the customer base is "New," against 80% "Loyal." Retention is working; growth now depends on top-of-funnel investment — paid acquisition, referral incentives, and welcome offers for first-time buyers.

### 3. Re-evaluate the subscription program

- Subscribers spend at essentially the same rate as non-subscribers, and most high-frequency repeat buyers are not subscribed. Redesign subscription perks to reward larger baskets (e.g., spend-based tiers, exclusive early access) and actively target the 2,518 non-subscribed repeat buyers for conversion.

### 4. Review markdown strategy on high-discount items

- Hats, sneakers, coats, sweaters, and pants are discounted on roughly half of all purchases. Assess whether this reflects necessary price sensitivity or margin erosion, and test smaller/targeted discounts or bundling instead of broad markdowns.

## **5. Promote express shipping**

- Express shipping orders average \$2 more than standard shipping. Test surfacing express shipping more prominently at checkout, or bundling it with loyalty perks, to lift average order value.

## **6. Feature top-rated and best-selling products**

- Highlight Gloves, Sandals, Boots, Hat, and Skirt (highest-rated) and the leading sellers in each category (Jewelry, Blouse, Sandals, Jacket, etc.) in marketing placements and homepage merchandising to capitalize on proven demand and satisfaction.

## **7. Sustain engagement across age groups**

- Revenue is evenly spread across age groups, with Young Adults slightly ahead. Maintain broad-based marketing rather than over-indexing on a single age segment, while testing incremental campaigns targeted at Young Adults to build on existing momentum.